



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-10	Team No- 06	Project Title- Grievance Redressal Portal for Institutions
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1. Abstract

The Grievance Redressal Portal for Institutions is an innovative, intelligent, and centralized platform designed to transform the traditional grievance-handling process into a transparent, trackable, and technology-driven system. The portal enables students, faculty, and staff to securely submit grievances related to academics, examinations, infrastructure, hostels, administration, and other institutional services.

Instead of relying on manual complaints and lengthy communication, the system provides real-time grievance tracking, automated categorization, priority-based assignment, status notifications, and role-based access. A combination of SQL and MongoDB is used to efficiently manage structured and flexible data, while a FastAPI based asynchronous REST backend provides secure APIs with JWT authentication, validation, and automated API documentation.

To improve scalability and reliability, the proposed system adopts a microservices architecture, allowing services such as authentication, grievance management, notifications, and administration to operate independently. REST APIs and Kafka-based asynchronous communication enable efficient interaction between services, while Docker provides consistent containerized deployment.

The system can also incorporate intelligent grievance classification and semantic search to help identify similar complaints and route grievances to the appropriate department. A CI/CD pipeline supports automated testing and deployment, making the platform suitable for real-world institutional environments. Overall, the project demonstrates how modern databases, intelligent automation, microservices, event-driven communication, containerization, and DevOps practices can be combined to create a scalable and accountable solution for institutional grievance management.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090005	M S S M V Sampath	
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3. Frameworks / Platforms / Software Used

- FastAPI – Asynchronous REST backend framework for building secure, well-documented APIs.
- SQL (PostgreSQL/MySQL) – Relational database for managing structured institutional data.
- MongoDB – NoSQL database for managing flexible, semi-structured grievance data.
- JWT (JSON Web Tokens) – Secure authentication and authorization for API access.
- Docker – Containerization platform for consistent, scalable service deployment.
- Apache Kafka – Event-streaming platform for asynchronous communication between microservices.
- Microservices Architecture – Design approach enabling independent, scalable services.
- REST APIs – Interface standard enabling communication between services and clients.
- CI/CD Pipeline – Automated testing and deployment pipeline for reliable releases.
- Postman – Tool for API testing and debugging.
- GitHub – Platform for version control and project collaboration.

Guide Name and Signature (remarks if any) 